

Final Report: PatchTST Experiments Summary

Overview

This document summarizes the PatchTST experiments carried out to predict crude oil price direction. Each experiment tested different model configurations including patch length, embedding dimension, number of layers, and dropout rates.

Throughout the experiments, the main goal was to achieve high directional accuracy, even at the expense of higher MAE or RMSE.

Important Observation

Despite changes in model architecture and hyperparameters, the predictions generated by the model consistently resulted in nearly constant values (a straight line). This behavior indicates that the model was not able to extract meaningful patterns or behaviors from the data. Directional predictions were only marginally better than random, with no clear response to real market fluctuations.

Experiment Results Summary

Exp#	Patch Len		Embed Dim		Layers	Dropout	Dir. Acc
19	10	256	2	0.1	50.00%		
24	10	128	3	0.1	50.97%		
33	10	128	3	0.1	51.95%		
41	25	128	3	0.1	51.65%		